

Algebra 1 – Q1.4_1.7 A

Name: _____

1.4 – Fractions, Decimals & Percents

1. I can convert between fractions, decimals, and percents

Convert as directed. Write answers in simplest form.

- a. $\frac{3}{4} \rightarrow$ decimal: _____ percent: _____
- b. $0.6 \rightarrow$ fraction: _____ percent: _____
- c. $45\% \rightarrow$ fraction: _____ decimal: _____
- d. $1.25 \rightarrow$ fraction: _____ percent: _____

2. I can decide if a fraction's decimal ends or repeats (and write it).

For each fraction: **write the decimal** (use a bar for repeating digits).

- a. $\frac{11}{25}$ — decimal: _____
- b. $\frac{7}{12}$ — decimal: _____
- c. $\frac{6}{20}$ — decimal: _____

3. I can compare and order fractions, decimals, and percents in real-world situations

Three stores are having a sale on coats with an original price of \$250. Here are the deals they offer.

- Store A: **\$100 off**
- Store B: **45% off**
- Store C: **$\frac{1}{2}$ off**

Which store gives the **biggest discount**? _____

1.5 – Multiply, Divide & Compare Fractions

4. I can multiply fractions and mixed numbers

Simplify your answers.

a. $\frac{3}{5} \times 10 = \underline{\hspace{2cm}}$

b. $\frac{2}{3} \times \frac{3}{4} = \underline{\hspace{2cm}}$

c. $1\frac{1}{2} \times 2\frac{2}{3} = \underline{\hspace{2cm}}$

5. I can divide fractions by using reciprocals

Find the following quotients.

a. $\frac{3}{4} \div \frac{2}{5} = \underline{\hspace{2cm}}$

b. $5 \div \frac{2}{3} = \underline{\hspace{2cm}}$

1.6 – Solving Problems with Fractions, Decimals & Percents

6. I can find a part, a whole, or a percent

- a. What is 30% of 80? _____
- b. 18 is 20% of what number? _____
- c. What percent is 45 out of 60? _____

7. I can use percents to solve problems involving grades, sales, and survey results

- a. A \$40 shirt is 25% off. After the discount, 8% sales tax is added. What is the final price? (Round to the nearest cent.)

Answer: _____

- b. In a survey of 120 students, 45% prefer soccer. How many students is that? _____

8. I can solve percent change problems in real-world contexts

A recipe uses $\frac{3}{4}$ cup of sugar. You reduce the sugar by 20%. How much sugar do you use now?

Answer: _____ cup(s)

1.7 – Order of Operations (PEMDAS)

9. I follow the correct order of operations (including fractions and negatives)

Simplify each expression.

a. $6 - \frac{3}{2} \cdot 4 =$ _____

b. $-3^2 + 5 =$ _____

c. $\frac{1}{2} + \frac{3}{4} \cdot 2 - 1 =$ _____

d. $\frac{8}{3} \div \left(\frac{2}{3}\right) - 1 =$ _____

10. I avoid common mistakes when simplifying expressions

A student says $4 \div \frac{1}{2} = 2$ “because you divide by 2.”

a. Explain what is wrong.

b. Find the correct value of $4 \div \frac{1}{2}$.

11. I avoid common mistakes with negatives and exponents

True or False? Explain your reasoning clearly.

“ $(-3)^2 = -9$ ”